FORMATIVE ASSIGNMENT

TULIA: A web-based emotional Literacy System for university students

- **Description of the concept of their Mission;**

    1. **Mission:**

    To support the mental wellbeing of early-stage university students in high-pressure academic environments by providing accessible, private and culturally sensitive digital tools, with a primary focus on higher education institutions in Africa.

    2. **Relevance in africa:**

    This mission is highly relevant in Africa, where mental health support for university students is often limited and stigmatized. Many universities lack adequate mental health services leaving early-stage students without timely support during critical academic transitions. A private and culturally sensitive digital approach reduces barriers to care and encourages early engagement. By focusing on emotional literacy rather than clinical treatment, the mission supports preventative mental wellbeing and helps students manage academic pressure more effectively.

- **Problem statement:**

Early-stage university students often experience emotional stress and anxiety as they adjust to academic pressure,independence and performance expectations. Research indicates that approximately 40% of university students report symptoms of anxiety or emotional distress, particularly during their initial years of study (WHO, 2023). Globally, over one-third of university students experience persistent stress symptoms, highlighting the widespread nature of emotional challenges in higher education (Patel et all., 2024). In African universities, limited mental health services and persistent stigma further restrict access to early support (UNESCO, 2023). Despite this prevalence, many students lack emotional literacy, reducing their ability to recognize and regulate emotions during high pressure academic periods such as

examinations. Existing digital mental health solutions such as ***BetterHelp*** and ***Headspace*** tend to focus on clinical intervention and mindful techniques. These approaches do not directly address the foundational skill gap of emotional literacy leaving a clear need for a private, non-clinical tool that helps students recognize, name and understand their own emotions before the issue escalates.

- **Software development model**
  I chose the Agile software development model for TULIA since the project will deal with a human-centered problem where user needs, emotional content and feature effectiveness may change as the system evolves. Agile helps me develop the system step by step, continuously review the development process and modify the features according to the feedback of users without interrupting the whole project.

  The way i will use this is by initially defining a MVP which concentrates on the core features. Development of the system will then be done in small cycles with each cycle being planned on what specific features are to be implemented, testing their functionality, reviewing feedback and refining the system before proceeding to the next cycle. This will ensure that the solution remains aligned with user requirements and project objectives throughout the development.

- **The hypothesis of their solution**
  I anticipate that once TULIA is deployed, early-stage university students who actively use the system will develop advanced emotional literacy demonstrated by increased emotional vocabulary, more consistent emotional check-ins and greater awareness of emotional patterns over time. Through the use of the structured check-ins and reflection features or the pattern awareness dashboard, users will have a better image of what is going on with their emotions and react more consciously in times of high pressure like their academics. Also, the accessibility of curated resources and helplines is expected to promote timely help-seeking when needed thereby maintaining healthier emotional coping behavior without the necessity of clinical intervention.

References

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Software Requirements Specification (SRS) Template

TULIA: A web-based emotional literacy system for University students

Prepared by Kellia Kamikazi

African Leadership University

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Revision History

Name	Date	Reason For Changes	Version
Kellia Kamikazi	25/01/2026	Completed SRS draft prepared for submission including all sections: purpose, scope, overview, functional and non-functional requirements	1.0

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for **TULIA**, a web-based emotional literacy system intended to support early-stage university students in managing emotional challenges in academic environments.This document applies to **TULIA version 1.0** which represents the initial release of the system. The requirements specified in this document define the expected behavior and quality attributes of the system requirements.

1.2 Document Conventions

This document uses clear and consistent structure to present software requirements. Headings and numbering are applied to organise content logically. Each requirement statement clearly explains what the system is expected to do. All requirements are considered equally important unless stated otherwise. The document avoids complex technical language to ensure that it can be easily understood by readers without a technical background.

1.3 Intended Audience and Reading Suggestions

This document is intended for university supervisors, project evaluators and readers who wish to understand the purpose and functionality of the TULIA system. It may also be used by individuals involved in reviewing the system. Readers seeking the general overview of the system should begin with the Introduction and Product scope sections whereas those who want more detailed information about system behavior can refer to the sections describing system requirements.

1.4 Product Scope

TULIA is a web-based emotional literacy system designed to help early-stage university students recognize, understand and reflect on their emotional experiences in academic settings. The system provides structured emotional check-ins, emotional vocabulary learning, personal reflections, pattern awareness and access to external resources and helplines. The primary objective of the system is to promote early emotional awareness and healthy coping behaviors through a private, non-clinical digital platform that complements existing student support services.

2. Overall Description

2.1 Product Perspective

TULIA is a new web-based system designed to support early-stage university students in managing emotional challenges in academic settings. It does not replace the existing student support but compliments them by providing a private, culturally sensitive and accessible platform for emotional literacy.

The system interacts with external resources such as helplines and other mental health resources. A high-level view of TULIA's components includes:

- **Web application interface**: for student interactions.
- **Backend server**: handles data storage, processing and analytics.
- **Database**: stores student inputs securely ensuring privacy and confidentiality.
- **External resource integration**: links to approved mental health resources and helplines

2.2 Product Functions

At a high level, TULIA enables students to:

- Log and track their daily emotions through structured check-ins
- Visualize emotional patterns and trends over time using a dashboard
- Learn and expand their emotional vocabulary to better understand and express their feelings
- Reflect on personal experiences and coping strategies
- Connect with culturally relevant external resources and mental health support

2.3 User Classes and Characteristics

Primary users: Early-stage university students

Characteristics:

- Students in their first or second year of university
- Basic digital literacy
- Regular access to internet-enabled devices
- Interested in managing emotions, self-reflection and improving wellbeing
- Require privacy and confidentiality when using the system

TULIA is designed exclusively for early-stage university students, providing them with a private, web-based platform to track emotions, reflect and improve emotional literacy without involving any external users in version 1.0.

2.4 Operating Environment

- Web browsers: Chrome, Firefox, Edge
- Devices: Laptops, tablets and smartphones
- Internet connection required
- Backend: cloud-hosted server supporting secure database storage
- Frontend technologies: HTML, CSS, Javascript

2.5 Design and Implementation Constraints

Privacy and data protection: TULIA must comply with relevant data protection regulations to ensure all student inputs, check-ins, reflection and activity data remain confidential.

Web-based platform: The system must operate within modern web browsers and be fully responsive on laptops, tablets and smartphones.

Cultural sensitivity: All content must be culturally appropriate and relevant to African university students.

Lightweight design: The application must be optimized to accommodate devices with limited processing power and varying internet speeds.

Technology stack constraints: Development should use open-source or approved technologies that support long-term maintainability.

Version 1.0 Scope: TULIA is designed for students only in the initial release; no administrative or counselors interfaces are included in this version.

2.6 User Documentation

- **User manual**: a comprehensive guide explaining all features, available in PDF for easy access.
- **Tutorial**: short step by step tutorials to introduce first-time users to the system.
- **FAQ section**: answers common questions about using the system, troubleshooting access issues and understanding privacy measures.

2.7 Assumptions and Dependencies

- **Internet access:** It's assumed that students will have access to the internet to use TULIA effectively
- **Device availability:** Students are expected to have access to a web-enabled device such as laptop, tablet or smartphone.
- **Digital literacy:** Users are assumed to have basic digital literacy including the ability to navigate web applications
- **Engagement:** It is assumed that students are willing to actively engage with the system's features
- **External resources:** The system depends on the availability of external mental health resources, helplines and culturally relevant content.
- **Browser compatibility:** TULIA assumes compatibility with modern browsers such as chrome, firefox,edge,... and that these browsers are up to date.
- **Scope limitation:** Version 1.0 of TULIA is designed exclusively for student use.

3. External Interface Requirements

3.1 User Interfaces

TULIA provides a web-based graphical user interface (GUI) designed for ease of use, accessibility and privacy. Key characteristics include:

- **Consistent layout:** Navigation menus, dashboards and feature screens follow a uniform design
- **Feature-specific screens:**

1) **Emotional check-ins:** simple forms to log emotions with optional free-text reflections
2) **Pattern awareness dashboard:** interactive chart and graphs to visualize emotional trends
3) **Reflections:** text input area linked to individual check-ins
4) **Vocabulary builder:** Interactive exercises with selectable answers and progress tracking
5) **External resources:** links displayed clearly with descriptions and accessibility labels

- **Error handling:** Clear error messages for invalid inputs, connectivity issues, etc
- **Accessibility:** The interface follows web accessibility guidelines (WCAG) for readability and usability

3.2 Hardware Interfaces

- **Supported devices:** laptops, tablets and smartphones with internet access
- **Data input:** keyboard,touch screen or mouse
- **Data output:** standard web display for forms ,charts and text areas
- **Hardware requirements:** devices must have sufficient processing power and memory to run modern web browsers; no additional hardware is needed
- **Network interface:** standard network activity (i.e: wi-fi or mobile data) is required to access the system

3.3 Software Interfaces

- **Web browsers:** Compatible with Chrome, Firefox, Edge and Safari (latest versions)
- **Backend server:** python-based server using flask to handle application logic and process student data
- **Database:** cloud-hosted relational database to securely store student check-ins, reflections and emotional data
- **Frontend:** HTML,CSS and Javascript for building interactive user interfaces
- **External resources:** Interfaces with mental health APIs using secure HTTP requests
- **Data exchange format:** JSON used for communication between frontend and backend
- **Libraries and tools:** python libraries for data handling, security (encryption) and visualization
- **Security & privacy:** All data transmitted over HTTPSand stored encrypted in the database to ensure privacy and integrity.

3.4 Communications Interfaces

- **Protocol:** All communications between the client (web browser) and the python backend server occurs over HTTPS to ensure secure data transfer.
- **Data format:** Data exchanged between frontend and backend is in JSON format
- **Real-time updates:** The system ensures that check-ins, reflections and dashboard visualizations update in real-time for the student
- **Security measures:** All communications are encrypted, sensitive information like login credentials is never transmitted in plain text
- **Error handling:** Communication failures, server downtime or data transmission errors trigger clear notifications to the user and automatic retry mechanisms where appropriate
- **Compatibility:** Communication standards are compatible with modern web browsers and do not require additional client-side plugins.

4. Requirement Specification

Functional Requirements

Req ID	Requirements	Description
FR 1	User authentication	The system shall allow university students to register and securely authenticate using login credentials
FR 2	Emotional check-ins	The system shall enable users to record emotional check-ins from a structured emotional vocabulary
FR 3	Reflection management	The system shall allow users to create, edit,view and delete personal emotions reflections
FR 4	Emotional vocabulary learning	The system shall provide definitions and examples of emotions to help users improve emotional literacy
FR 5	Pattern awareness dashboard	The system shall analyze check-in data and display visual summaries of emotional pattern over time
FR 6	External support resources	The system shall provide access to curated mental health resources and helplines to university students in Africa
FR 7	Data storage and retrieval	The system shall securely store and retrieve user emotional data including check-ins and reflections
FR 8	User data deletion	The system shall allow users to permanently delete their account and associated emotional data upon request

5. Other Nonfunctional Requirements

Requirement Type	Req ID	Description
Usability	NFR 1	The system shall be easy to use and understandable by university students with basic digital literacy
Performance	NFR 2	The system shall respond to user requests within 3 seconds under normal operating conditions
Availability	NFR 3	The system shall be available 24/7 except during scheduled maintenance periods
Security	NFR 4	The system shall encrypt sensitive user data during transmission and while stored in the database
Scalability	NFR 5	The system shall support growth in the number of users without major changes to system architecture
Privacy	NFR 6	The system shall ensure that emotional is accessible only to the authenticated user
Compatibility	NFR 7	The system shall operate on modern web browsers across desktop and mobile devices
Maintainability	NFR 8	The system shall be modular and well documented to allow easy updates and future enhancements

Guide to Non- Functional Requirements

Performance Requirements

This system is designed to be responsive and support frequent student use. Dashboards and check-ins shall load quickly to ensure smooth interactions even with multiple concurrent users. This performance ensures students can engage consistently during high-pressure academic periods.

Safety Requirements

TULIA is a non-clinical tool that promotes emotional literacy and awareness. Its safety measures include the use of supportive non-diagnostic language, clear disclaimers that it is not a substitute for

professional mental health services and access to external emergency resources. Unauthorized access to user data is prevented to avoid emotional or reputational harm.

Security Requirements

To protect user privacy, TULIA enforces authentication for all accounts, encrypts sensitive data in transit and at rest and restricts access to only the authenticated user. The system also adheres to applicable data protection and privacy guidelines for higher education institutions in Africa.

Software Quality Attributes

TULIA prioritizes:

Usability: easy for students with basic digital literacy

Reliability & Availability: consistent operation with minimal downtime

Maintainability & Scalability: modular design that allows updates and growth

Portability &compatibility: works on modern browsers and mobile/desktop devices

Testability: requirements are verifiable and measurable

Business Rules

- Only registered students may engage with the system
- User cannot view other students' emotional data
- Users can delete their own data at any time
- External mental health resources are available to all users without requiring data submission

6. Appendix

Appendix A: Glossary

TULIA – Web-based emotional literacy system designed to support early-stage university students in understanding and managing their emotions in academic environments.

SRS – Software requirements specification

GUI – Graphical user interface

WSAG – Web content accessibility guidelines

MVP – Minimum viable product

WHO – World health organization

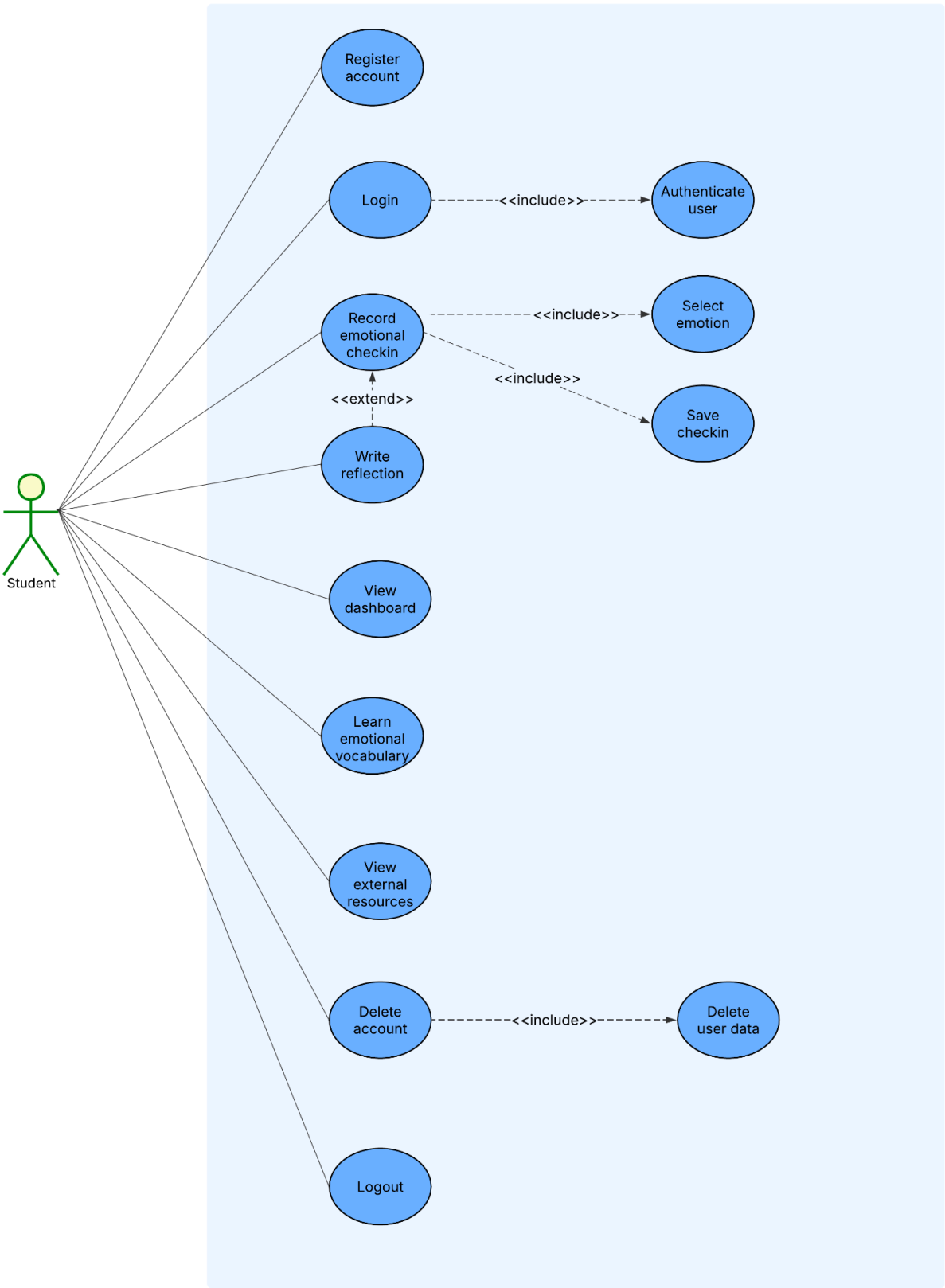
UNESCO – United Nations Educational Scientific and Cultural Organization

Appendix B: Analysis models

I. Use Case Diagram

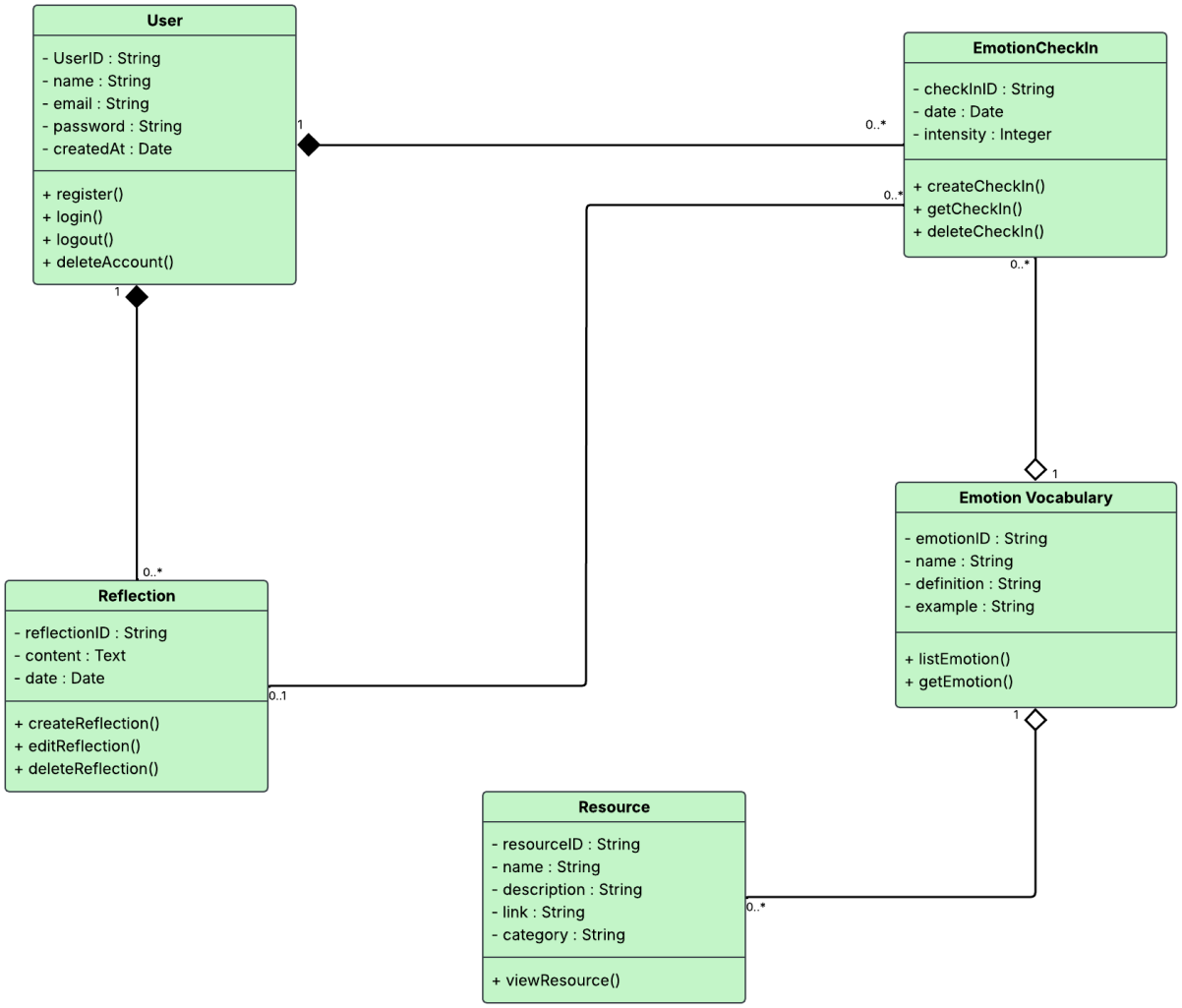
This diagram focuses on the User actor and the core system functionalities such as authentication, emotion tracking, reflection writing, and accessing support resources. It highlights relationships like «include» and «extend», showing how some actions depend on others (e.g: login required before creating a check-in)

TULIA: Emotional Literacy System (V1.0)



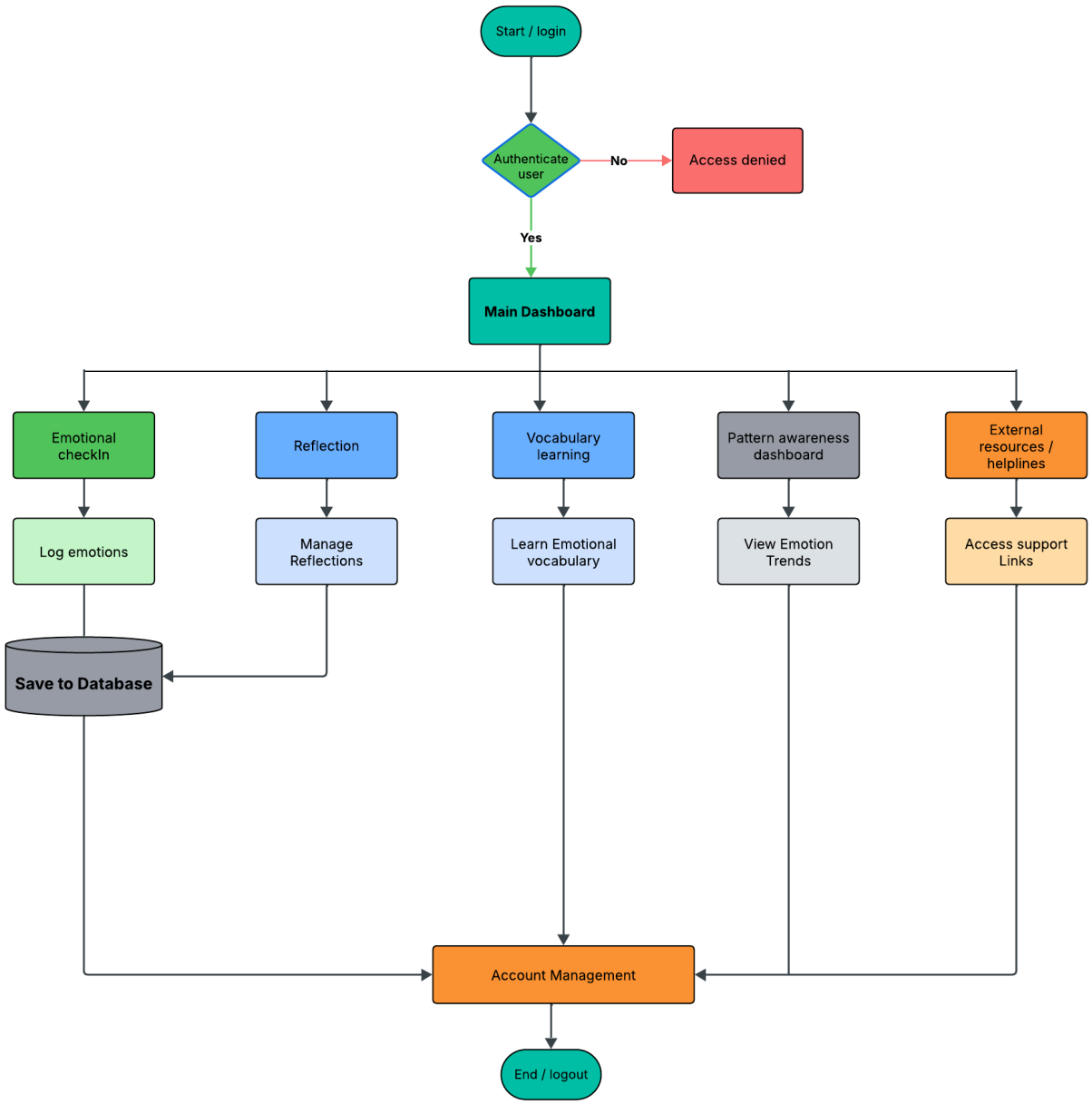
II. Class Diagram

This diagram emphasizes on the major system classes: User, EmotionCheckIn, Reflection, EmotionVocabulary and Resource. It shows how User composes check-ins and reflections, how emotions are categorized and how different entities are connected through association, aggregation and composition relationships



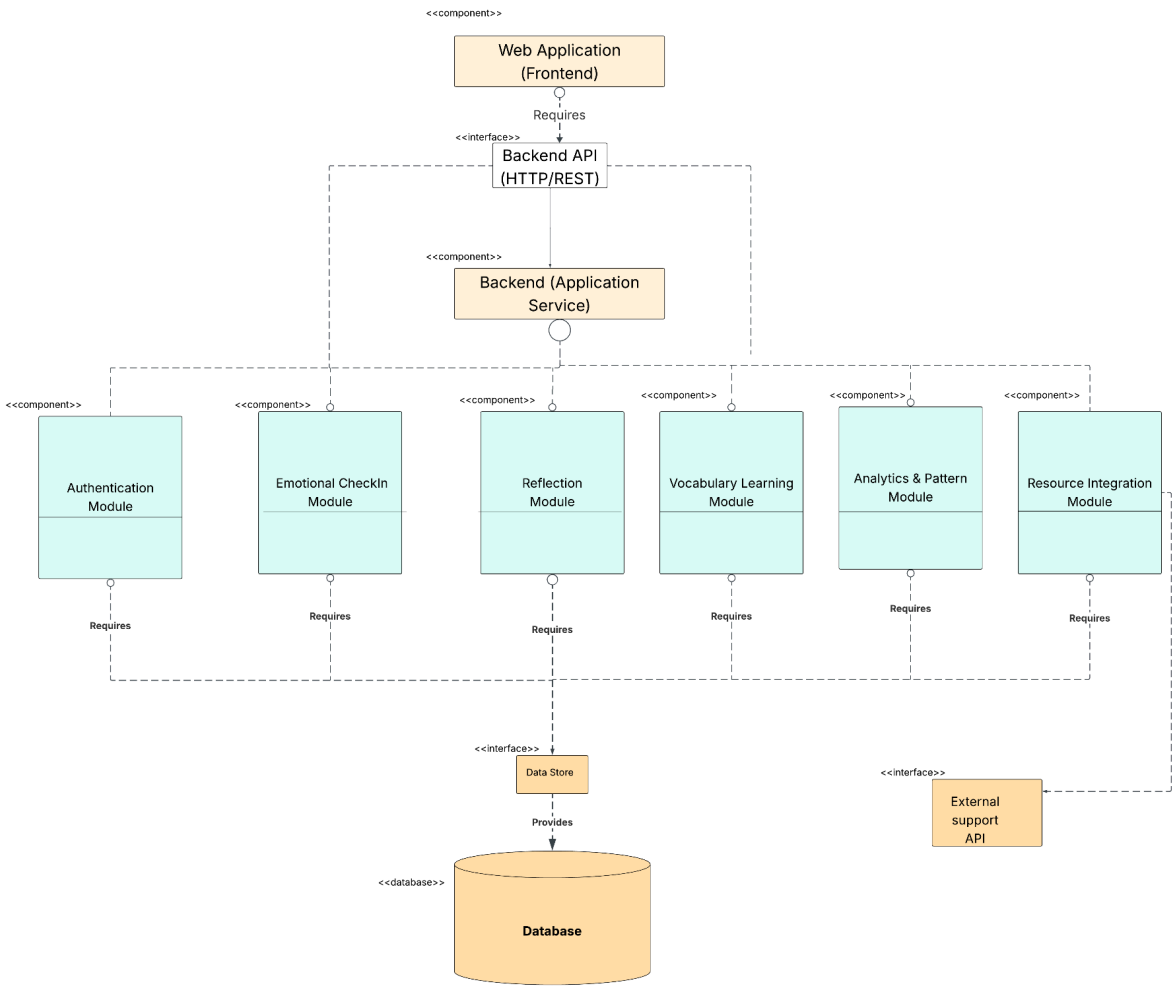
III. Flowchart Diagram

This flowchart represents the step-by-step process of how a user interacts with the TULIA. It begins with login and authentication, then directs the user to the main dashboard where they can perform actions such as emotional check-ins, reflections, vocabulary learning, viewing trends, and accessing resources. The flow shows how data is saved to the database and ends with account management and logout. The main focus is on the system's process flow and decision points (e.g: authentication approval or denial)



IV. Component Diagram

Below is the component diagram that highlights key system components including the User Interface, Authentication, Emotion Management, Reflection Management and Database. It explains how each component has a specific responsibility and how they interact to process and store user data.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Web Application (Frontend)
Requires
<<interface>>
Backend API (HTTP/REST)
<<component>>
Backend (Application Service)
<<component>>
Authentication Module
<<component>>
Emotional CheckIn Module
<<component>>
Reflection Module
<<component>>
Vocabulary Learning Module
<<component>>
Analytics & Pattern Module
<<component>>
Resource Integration Module
Requires
Requires
Requires
Requires
Requires
Requires
<<interface>>
Data Store
Provides
<<interface>>
External support API
<<database>>
Database